

BCS303 Discrete Structures & Theory of Logic		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to understand		
CO 1	Acquire Knowledge of sets and relations for solving the problems of POSET and lattices.	K ₃ , K ₄
CO 2	Apply fundamental concepts of functions and Boolean algebra for solving the problems of logical abilities.	K ₁ , K ₂
CO 3	Employ the rules of propositions and predicate logic to solve the complex and logical problems.	K ₃
CO 4	Explore the concepts of group theory and their applications for solving the advance technological problems.	K ₁ , K ₄
CO 5	Illustrate the principles and concepts of graph theory for solving problems related to computer science.	K ₂ , K ₆
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Set Theory & Relations: Introduction, Combination of sets. Relations: Definition, Operations on relations, Properties of relations, Composite Relations, Equality of relations, Recursive definition of relation, Order of relations. POSET & Lattices: Hasse Diagram, POSET, Definition & Properties of lattices – Bounded, Complemented, Distributed, Modular and Complete lattice.	08
II	Functions: Definition, Classification of functions, Operations on functions. Growth of Functions. Boolean Algebra: Introduction, Axioms and Theorems of Boolean algebra, Algebraic manipulation of Boolean expressions. Simplification of Boolean Functions, Karnaugh maps.	08
III	Theory of Logics: Proposition, Truth tables, Tautology, Satisfiability, Contradiction, Algebra of proposition, Theory of Inference. Predicate Logic: First order predicate, well-formed formula of predicate, quantifiers, Inference theory of predicate logic.	08
IV	Algebraic Structures: Definition, Groups, Subgroups and order, Cyclic Groups, Cosets, Lagrange's theorem, Normal Subgroups, Permutation and Symmetric groups, Group Homomorphisms, Definition and elementary properties of Rings and Fields.	08
V	Graphs: Definition and terminology, Representation of graphs, Multigraphs, Bipartite graphs, Planar graphs, Isomorphism and Homeomorphism of graphs, Euler and Hamiltonian paths, Graph coloring. Combinatorics: Introduction, Counting Techniques, Pigeonhole Principle	08

Text books:

1. Koshy, Discrete Structures, Elsevier Pub. 2008 Kenneth H. Rosen, Discrete Mathematics and Its Applications, 6/e, McGraw-Hill, 2006.
2. B. Kolman, R.C. Busby, and S.C. Ross, Discrete Mathematical Structures, 5/e, Prentice Hall, 2004.
3. E.R. Scheinerman, Mathematics: A Discrete Introduction, Brooks/Cole, 2000.
4. R.P. Grimaldi, Discrete and Combinatorial Mathematics, 5/e, Addison Wesley, 2004
5. Lipschutz, Seymour, "Discrete Mathematics", McGraw Hill.
6. Trembley, J.P & R. Manohar, "Discrete Mathematical Structure with Application to Computer Science", McGraw Hill.
4. Deo, Narsingh, "Graph Theory With application to Engineering and Computer Science.", PHI.
8. Krishnamurthy, V., "Combinatorics Theory & Application", East-West Press Pvt. Ltd., New Delhi

Set :

- A set is any well-defined collection of objects, called the elements or members of the set.
- These elements may be anything : numbers, points in geometry, letters of alphabets, etc.
- Capital letters $A, B, C, \dots$ are ordinarily used to denote sets and lowercase letters $a, b, c, \dots$ to denote elements of sets.
- The statement 'x is an element of A' or equivalently 'x belongs to A' is written as $x \in A$.
- The statement 'x is not an element of A' is written as $x \notin A$.

Representation of a set :

1. Roster or Tabular form
2. Rule method or Set builder form

Roster or Tabular form :

- In this form all the elements of the set are listed, the elements being separated by commas and are enclosed within braces.
- For ex., $A = \{a, e, i, o, u\}$.

Note :

- The order in which the elements of a set are listed is not important. Thus $\{1, 2, 3\}$, $\{1, 3, 2\}$, $\{2, 3, 1\}$ are all representation of same set.

Null set:

- A set which contains no elements is called a null set.

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- Repeated elements in the listing of the elements of a set can be ignored. Thus $\{1, 3, 2, 3, 1\}$ is another representation of the set $\{1, 2, 3\}$.

Rule Method or Set builder form:

- In this method, a set is defined by specifying a property that elements of the set have in common.
- The set is then described as follows:

$$A = \{x : p(x)\}$$

A vertical bar is also used in place of colon (:) in the set builder form.

- For ex, $A = \{x : x \text{ is a vowel in the English alphabet}\}$

Note:

- In particular, some sets are represented by special symbols. This is

$$N = \{x : x \text{ is a natural number}\}$$

$$Z = \{x : x \text{ is an integer}\}$$

$$R = \{x : x \text{ is a real number}\}$$

Finite set:

- A set with finite number of elements in it, is called a finite set.
- For ex, The set of months in a year.

Infinite set:

- An infinite set is a set which contains infinite number of elements.
- For ex. Set of integers.

Null set:

- A set which contains no elements at all is called the Null set.
- Also known as Empty set or Void set.
- It is denoted by the symbol ϕ , $\{\}$.
- For ex, $A = \{x: x^2 + 4 = 0, x \text{ real}\}$

Singleton set:

- A set which has only one element is called a singleton set.
- For ex, $S = \{a\}$ is a singleton set.

Subset:

- If A and B are sets such that every element of A is also an element of B , then A is said to be a subset of B , (or A is contained in B), and is denoted by $A \subseteq B$.
- In other words,
$$A \subseteq B, \text{ if } x \in A \text{ and } x \in B.$$
- If A is not a subset of B , i.e., at least one element of A does not belong to B , we write $A \not\subseteq B$.
- The following results are very important.
 1. Every set A is a subset of itself, i.e., $A \subseteq A$.
 2. The null set ϕ is considered as a subset of any set A i.e., $\phi \subseteq A$.

- It is written as $A \subseteq B$.

- For ex, if $A = \{1, 5\}$, $B = \{1, 5, 6\}$, $C = \{1, 6, 5\}$

then A and B are both subsets of C ; but A is a proper subset of C , whereas B is not. (4)

3. If A is a subset of B , and B is a subset of C , then A is a subset of C , i.e.,
if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

- For ex, the set $A = \{1, 3, 4\}$ is a subset of $B = \{1, 2, 3, 4, 5\}$ because each element of A also belongs to the set B .

Number of Subsets of a set:

- If a set contains n elements, then the number of subsets is 2^n .

- For ex, set $A = \{2, 4, 5, 6, 7\}$ contains five elements, so that total number of its subsets $= 2^5 = 32$.

Ex: List all the subsets of the set $A = \{a, b, c\}$

→ The given set contains three elements, hence it has $2^3 = 8$ subsets. The subsets are ϕ , $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{a, b, c\}$.

Super set:

- If A is a subset of B , then B is called the super set of A and is written as $B \supseteq A$ which is read as 'B is a super set of A'.

Proper subset:

- Any subset A is said to be proper subset of another set B if A is a subset of B , but there is at least one element of B which does not belong to A .

- It is written as $A \subset B$.

- For ex; if $A = \{1, 5\}$, $B = \{1, 5, 6\}$, $C = \{1, 6, 5\}$

then A and B are both subsets of C ; but A is a proper subset of C , whereas B is not a proper subset of C since $B = C$.

Equal set:

- Two sets A and B are said to be equal if and only if every element of A is an element of B and consequently every element of B is an element of A ; that is $A \subset B$ and $B \subset A$ and it is written as $A = B$.

- Symbolically, $A = B$ if $x \in A \& x \in B$.

Universal set:

- A non-empty set of which all the sets under consideration are subsets is called the universal set.

- It is also called as Universe of Discourse set and is denoted by U .

- For ex; In a study of human population, all people in the world may be assumed to form the Universal set. The people of any continent, country, religion is a subset of this universal set.

Operations on sets:

Union:

- The union of two sets A and B , denoted by $A \cup B$, pronounced as 'A union B' is the set of all elements which belong to A or to B ; that is,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

do not belong to B; that is,

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Intersection :

- The intersection of two sets A and B, denoted by $A \cap B$, pronounced as 'A intersection B', is the set of elements which belong to both A and B; that is,

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

- If $A \cap B = \phi$, that is, if A and B do not have any elements in common, then A and B are said to be disjoint or non-intersecting.

Ex: Let $A = \{1, 2, 3, 4\}$, $B = \{0, 1, 3, 5, 7\}$ and $C = \{2, 4, 6, 8\}$,

then $A \cup B = \{0, 1, 2, 3, 4, 5, 7\}$ $A \cap B = \{1, 3\}$

$$B \cup C = \{0, 1, 2, 3, 4, 5, 6, 7, 8\} \quad B \cap C = \phi$$

Complements :

- Let U be the universal set and A be any subset of U. The absolute complement of A or, simply, complement of A, denoted by A' or A^c is the set of elements which belong to U but which do not belong to A; that is,

$$A' = \{x : x \in U \text{ and } x \notin A\}$$

Ex:

Let $N = \{1, 2, 3, \dots\}$ be the universal set and $A = \{1, 3, 5, \dots\}$, then $A' = \{2, 4, 6, \dots\}$

Difference :

- If A and B are two sets, the relative complement of B with respect to A or, simply, the difference of A and B, denoted by $A - B$, is the set of elements...

which do not belong to B; that is,

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

Ex:

Let $A = \{a, b, c\}$ and $B = \{b, c, d, e\}$

then $A - B = \{a\}$ and

$$B - A = \{d, e\}$$

Symmetric Difference:

- The symmetric difference of two sets A and B, denoted by $A \Delta B$ or $A \oplus B$ is the set of elements that belong to A or to B, but not to both A and B.
- It is also called the Boolean sum of two sets.
- It is easy to see that

$$\begin{aligned} A \Delta B &= (A - B) \cup (B - A) \text{ or } (A \cup B) - (A \cap B) \\ &= \{x : x \text{ belongs to exactly one of A and B}\} \end{aligned}$$

Ex:

If $A = \{-3, 0, 1, 2\}$ and $B = \{1, 2, 3, 4\}$,

then $A - B = \{-3, 0\}$ and $B - A = \{3, 4\}$

$$\begin{aligned} \text{So, } A \Delta B &= \{-3, 0\} \cup \{3, 4\} \\ &= \{-3, 0, 3, 4\} \end{aligned}$$

Algebra of sets:

Idempotent laws:

1) a) $A \cup A = A$

b) $A \cap A = A$

Associative laws:

2) a) $(A \cup B) \cup C = A \cup (B \cup C)$

b) $(A \cap B) \cap C = A \cap (B \cap C)$

Let x be an element of $A \cup B$. Then -

Commutative laws:

$$\begin{aligned} 3) \quad a) \quad A \cup B &= B \cup A \\ b) \quad A \cap B &= B \cap A \end{aligned}$$

Distributive laws:

$$\begin{aligned} 4) \quad a) \quad A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \\ b) \quad A \cap (B \cup C) &= (A \cap B) \cup (A \cap C) \end{aligned}$$

Identity laws:

$$\begin{aligned} 5) \quad a) \quad A \cup \phi &= A \\ b) \quad A \cap U &= A \end{aligned}$$

$$\begin{aligned} 6) \quad a) \quad A \cup U &= U \\ b) \quad A \cap \phi &= \phi \end{aligned}$$

Involution law:

$$7) \quad (A')' = A$$

Complement laws:

$$\begin{aligned} 8) \quad a) \quad A \cup A' &= U \\ b) \quad A \cap A' &= \phi \end{aligned}$$

$$\begin{aligned} 9) \quad a) \quad U' &= \phi \\ b) \quad \phi' &= U \end{aligned}$$

De Morgan's laws:

$$\begin{aligned} 10) \quad a) \quad (A \cup B)' &= A' \cap B' \\ b) \quad (A \cap B)' &= A' \cup B' \end{aligned}$$

Proof: 1) (a) $A \cup A = A$

→ Let x be any arbitrary element of the set A , then

$$\begin{aligned} x \in A &\Rightarrow x \in A \text{ or } x \in A \\ &\Rightarrow x \in A \cup A \end{aligned}$$

Thus, $A \subseteq A \cup A$ ————— (i)

Conversely, if $x \in A \cup A$, then $x \in A$ or $x \in A$

Hence, $A \cup A \subseteq A$ ————— (ii)

From, (i) and (ii), $A \cup A = A$ (proved)

Proof: 3(a) $A \cup B = B \cup A$

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→ Let x be an element of $A \cup B$. Then

$$\begin{aligned} x \in A \cup B &\Rightarrow x \in A \text{ or } x \in B \\ &\Rightarrow x \in B \text{ or } x \in A \\ &\Rightarrow x \in B \cup A \end{aligned}$$

Thus, $A \cup B \subseteq B \cup A$ ————— (i)

Conversely, let x be any element of $B \cup A$, then

$$\begin{aligned} x \in B \cup A &\Rightarrow x \in B \text{ or } x \in A \\ &\Rightarrow x \in A \text{ or } x \in B \\ &\Rightarrow x \in A \cup B \end{aligned}$$

Thus, $B \cup A \subseteq A \cup B$ ————— (ii)

From (i) and (ii), $A \cup B = B \cup A$ (proved)

Proof: 3(b) $A \cap B = B \cap A$

→ Let x be any element of $A \cap B$, then

$$\begin{aligned} x \in A \cap B &\Rightarrow x \in A \text{ and } x \in B \\ &\Rightarrow x \in B \text{ and } x \in A \\ &\Rightarrow x \in B \cap A \end{aligned}$$

Thus, $A \cap B \subseteq B \cap A$ ————— (i)

Conversely, let x be any element of $B \cap A$, then

$$\begin{aligned} x \in B \cap A &\Rightarrow x \in B \text{ and } x \in A \\ &\Rightarrow x \in A \text{ and } x \in B \\ &\Rightarrow x \in A \cap B \end{aligned}$$

Thus, $B \cap A \subseteq A \cap B$ ————— (ii)

From (i) and (ii), $A \cap B = B \cap A$ (proved)



Then, $x \in A' \cup B' \Rightarrow x \in A' \text{ or } x \in B'$ (11)

Proof: 4(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (10)

$\longrightarrow$ Let x be any arbitrary element of the set $A \cup (B \cap C)$.
Then, $x \in A \cup (B \cap C) \Rightarrow x \in A \text{ or } x \in (B \cap C)$
 $\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C)$
 $\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$
 $\Rightarrow x \in A \cup B \text{ and } x \in A \cup C$
 $\Rightarrow x \in (A \cup B) \cap (A \cup C)$

Thus, $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \text{ — (i)}$

Conversely, let x be any arbitrary element of the set $(A \cup B) \cap (A \cup C)$

Then, $x \in (A \cup B) \cap (A \cup C) \Rightarrow x \in A \cup B \text{ and } x \in A \cup C$
 $\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C)$
 $\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C)$
 $\Rightarrow x \in A \text{ or } x \in B \cap C$
 $\Rightarrow x \in A \cup (B \cap C)$

Thus, $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \text{ — (ii)}$

From, (i) and (ii), $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (proved)

Proof: 10(b) $(A \cap B)' = A' \cup B'$

$\longrightarrow$ Let x be any arbitrary element of the set $(A \cap B)'$.

Then, $x \in (A \cap B)' \Rightarrow x \notin (A \cap B)$
 $\Rightarrow x \notin A \text{ or } x \notin B$
 $\Rightarrow x \in A' \text{ or } x \in B'$
 $\Rightarrow x \in (A' \cup B')$

Thus, $(A \cap B)' \subseteq (A' \cup B') \text{ — (i)}$

Conversely, let x be any arbitrary element of the set $A' \cup B'$.

$$\text{Then, } x \in A' \cup B' \Rightarrow x \in A' \text{ or } x \in B'$$

$$\Rightarrow x \notin A \text{ or } x \notin B$$

$$\Rightarrow x \text{ does not belong to both } A \text{ and } B$$

$$\Rightarrow x \notin (A \cap B)$$

$$\Rightarrow x \in (A \cap B)'$$

$$\text{Thus, } A' \cup B' \subseteq (A \cap B)' \quad \text{--- (i)}$$

$$\text{From (i) and (ii), } (A \cap B)' = A' \cup B' \quad \text{(proved)}$$

Cardinal Number in a Set :

- The cardinal number of a set A is the number of elements in the set A .

- It is denoted by $|A|$ or $n(A)$.

Properties :

1) For any two sets A and B

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

If A and B are disjoint, i.e., if $A \cap B = \phi$, then

$$n(A \cup B) = n(A) + n(B)$$

$$2) \quad n(A - B) = n(A) - n(A \cap B)$$

$$n(B - A) = n(B) - n(A \cap B)$$

3) If A, B, C are any three sets, then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

The multisets A and B can also be written as

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Power Set:

- If S is any set, then the family of all the subsets of S is called the power set of S .
- The power set of S is denoted by $P(S)$. Symbolically,
$$P(S) = \{T : T \subseteq S\}$$
- Obviously, ϕ and S are both elements of $P(S)$.
- If the set S is finite and contain n elements, then the power set of S will contain 2^n elements.

Ex:

i) If $A = \{1, 2\}$, then $P(A) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$

ii) If $B = \{a, b, c\}$, then $P(B) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$

Multiset:

- In specifying a set by its elements, it makes no difference how many times a particular element is repeated. Sometimes, however, the frequency with which a particular element appears in a list may be relevant. For ex; when we consider the set of marks obtained by the students in a class at some examination, we would like to count each figure as many times as the number of students who score it. The appropriate mathematical concept to handle such situations is a multiset.

- Multisets are sets where an element can occur as a member more than once.

- For ex., $A = \{a, a, a, b, b, c\}$, $B = \{a, a, a, a, b, b, b, d, d\}$ (13)
are multisets.

- The multisets A and B can also be written as

$$A = \{3 \cdot a, 2 \cdot b, 1 \cdot c\} \quad \text{and} \quad B = \{4 \cdot a, 3 \cdot b, 2 \cdot d\}$$

- The multiplicity of an element in a multiset is defined to be the number of times the element appears in the multiset.

- The cardinality of a multiset is defined to be the cardinality of the set it corresponds to, assuming that the elements in the multiset are all distinct.

Let A and B be multisets.

- The union of A and B , denoted by $A \cup B$, is the multiset where the multiplicity of an element is the maximum of its multiplicities in A and B .

- The intersection of A and B , denoted by $A \cap B$, is the multiset where the multiplicity of an element is the minimum of its multiplicities in A and B .

- The difference of A and B , denoted by $A - B$, is the multiset where the multiplicity of an element in A minus the multiplicity of the element in B if the difference is positive, and is equal to zero if the difference is 0 and negative.

- The sum of A and B , denoted by $A+B$, is the multiset where the multiplicity of an element is the sum of multiplicities of the element in A and B . (14)

Ex:

Let P and Q be two multisets $\{3 \cdot a, 2 \cdot b, 1 \cdot c\}$ and $\{4 \cdot a, 3 \cdot b, 2 \cdot d\}$ respectively. Find

- (a) $P \cup Q$ (b) $P \cap Q$ (c) $P - Q$ (d) $Q - P$ (e) $P + Q$

→

(a) $P \cup Q = \{4 \cdot a, 3 \cdot b, 1 \cdot c, 2 \cdot d\}$

(b) $P \cap Q = \{3 \cdot a, 2 \cdot b\}$

(c) $P - Q = \{1 \cdot c\}$

(d) $Q - P = \{1 \cdot a, 1 \cdot b, 2 \cdot d\}$

(e) $P + Q = \{7 \cdot a, 5 \cdot b, 1 \cdot c, 2 \cdot d\}$

Ex:

Let P and Q be two multisets where $P = \{5 \cdot a, 7 \cdot b, 8 \cdot c\}$ and $Q = \{3 \cdot a, 5 \cdot b, 9 \cdot c\}$. Find

- (i) $P \cup Q$ (ii) $P \cap Q$ (iii) $P - Q$ (iv) $Q - P$ (v) $P + Q$

→

(i) $P \cup Q = \{5 \cdot a, 7 \cdot b, 9 \cdot c\}$

(ii) $P \cap Q = \{3 \cdot a, 5 \cdot b, 8 \cdot c\}$

(iii) $P - Q = \{2 \cdot a, 2 \cdot b\}$

(iv) $Q - P = \{1 \cdot c\}$

(v) $P + Q = \{8 \cdot a, 12 \cdot b, 17 \cdot c\}$

Ordered Pairs and Cartesian Product:

Ordered Pair :

- An ordered pair is a pair of objects whose components occur in a special order.
- It is written by listing the two components in the specified order, separating them by a comma and enclosing the pair in parenthesis.
- In the ordered pair (a, b) , a is called the first component and b , the second component.

Cartesian product:

- Let A and B be sets. Cartesian product of A and B denoted by $A \times B$, is defined as

$$A \times B = \{(a, b) : a \in A, \text{ and } b \in B\}$$

i.e., $A \times B$ is the set of all possible ordered pairs whose first component comes from A and whose second component comes from B .

- For ex., if $A = \{a, b\}$ and $B = \{1\}$, then

$$A \times B = \{(a, 1), (b, 1)\}$$

$$B \times A = \{(1, a), (1, b)\}$$

$$A \times A = \{(a, a), (a, b), (b, a), (b, b)\}$$

Note:

- The cartesian product $A \times B \neq B \times A$. In $A \times B$, the elements of A will appear as the first component of the ordered pairs and in $B \times A$, the elements of B will appear

Let A and B be two sets as follows:

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as the first component of the ordered pairs.

Properties of Cartesian Product:

For the four sets A, B, C and D

$$1. (A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$$

$$2. (A - B) \times C = (A \times C) - (B \times C)$$

$$3. (A \cup B) \times C = (A \times C) \cup (B \times C)$$

$$4. A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Proof: (4) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

→ Let (x, y) be any element of $A \times (B \cap C)$.

$$\text{Then, } (x, y) \in A \times (B \cap C) \Rightarrow x \in A, \text{ and } y \in (B \cap C)$$

$$\Rightarrow x \in A, \text{ and } (y \in B \text{ and } y \in C)$$

$$\Rightarrow (x \in A, \text{ and } y \in B) \text{ and } (x \in A, \text{ and } y \in C)$$

$$\Rightarrow (x, y) \in A \times B \text{ and } (x, y) \in A \times C$$

$$\Rightarrow (x, y) \in (A \times B) \cap (A \times C)$$

$$\text{Thus, } A \times (B \cap C) \subseteq (A \times B) \cap (A \times C) \text{ — — (i)}$$

$$\text{Again, } (x, y) \in (A \times B) \cap (A \times C) \Rightarrow (x, y) \in A \times B \text{ and } (x, y) \in A \times C$$

$$\Rightarrow (x \in A, y \in B) \text{ and } (x \in A, y \in C)$$

$$\Rightarrow x \in A, (y \in B \text{ and } y \in C)$$

$$\Rightarrow x \in A, y \in (B \cap C)$$

$$\Rightarrow (x, y) \in A \times (B \cap C)$$

$$\text{Thus, } (A \times B) \cap (A \times C) \subseteq A \times (B \cap C) \text{ — — (ii)}$$

From (i) and (ii), we get,

$$A \times (B \cap C) = (A \times B) \cap (A \times C) \text{ (Proved)}$$